MIL-B-60992A (MU)
10 December 1971
SUPERSEDING
MIL-B-60992 (MU)
19 February 1968

MILITARY SPECIFICATION

BOMB, GENERAL PURPOSE, 750 POUND, M117A3
PARTS FOR

1. SCOPE

1.1 This specification covers special requirements not normally required by the drawings and provides quality assurance provisions for the fabrication of parts and the assembly of metal parts assemblies designated as: Bomb, General Purpose, M117A2 for Loading with Tritonal as the explosive charge.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

TT-C-490 - Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings.
MIL-A-2550 - Ammunition and Special Weapons, General Specification for.
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of.
MIL-C-45662 - Calibration of Standards.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Quality Assurance Terms and Definitions.
MIL-STD-248 - Qualification Tests for Welders; Other than Aircraft Weldments, Navy.

FSC 1325

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MIL-STD-1167 - Ammunition Data Cards.
MIL-STD-1168 - Lot Numbering of Ammunition.
MIL-STD-1235 - Single and Mutilevel Continuous Inspection by Attributes.

FEDERAL

Fed. Test Method Standard No. 151 - Metals, Test Methods.

DRAWINGS

ARMY

9226252 - Bomb, General Purpose, M117A3, Metal Parts Assembly.
9226253 - Body Assembly

PUBLICATIONS

ARMY

ET 9226252 - Equipment Tabulations.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Material.-Material and parts shall be in accordance with applicable specifications and drawings.

3.2 Bomb parts assembly.-The bomb parts assembly shall comply will all requirements specified on Drawing (dwg.) 9226252 and with all requirements specified in applicable specifications.

3.3 Strength of fuze well assembly welds. The fuze well assembly shall not become disassembled, shall not exhibit cracked welds, and shall not rupture or deform beyond drawing requirements after a load of 8,000 pounds, minimum, has been applied to each weld in a manner to effect disassembly.

3.4 Internal hydrostatic of body assembly.-The body assembly, after all welding and heat treatment, shall not rupture or deform in excess of drawing dimensions nor leak due to incomplete welds, voids or porosity, when subjected to an internal hydrostatic pressure of 2664 pounds per square inch, (p.s.i.), minimum, when tested as specified in 4.4.2.

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3.5 Qualification of welders.-Prior to use in production, all welders shall be qualified in accordance with Standard MIL-STD-248.

3.6 Welds.

3.6.1 Weld surface defects.-Welds shall be reasonably smooth and free of cracks, porosity, and craters. Welds shall be of uniform width, and height. They shall be free of irregularities which indicate lack of skill or experience on the part of the operator. All scale and weld splatter shall be completely removed prior to application of protective coating.

3.6.2 Base metal.-The heat-affected zone of the base metal shall be free from surface cracks. No undercutting of the base metal shall be permitted.

3.6.3 Integrity of circumferential tail weld.-Subsequent to all heat treatment operations and hydro-test the circumferential tail weld shall not contain any single discontinuity in excess of one (1) inch nor shall the sum of all the discontinuities exceed six (6) inches. A minimum of three (3) inches of sound weld shall separate discontinuities of 3/4 inch or more. Discontinuities of 1/8 inch and smaller shall be disregarded. Discontinuities between 1/8 inch and 3/4 inch in length, when separated by less than 3/4 inch of sound weld shall be considered as one continuous discontinuity.

3.7 Surface preparation prior to painting.-All painted surfaces shall be prepared for painting in accordance with Method I, Type I of Specification TT-C-490.

3.8 First article testing.-Requirement for the submission of first article samples by the contractor shall be as specified in the contract.

3.9 Workmanship.-All parts shall be free of chips, dirt, grease, rust, and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents.

3.10 Stress crack test.-The shipping protector ring shall show no evidence of cracks or breaks when tested as specified in 4.4.5.

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3.11 Seal of fuzing compartments.-The fuzing compartments consist of both fuze well assemblies and conduits. The joints between the conduits and the charging well, the joint between the conduit and each fuze well assembly and both fuze well assemblies, in the assembled parts assembly, shall be sealed to the extent to assure against leakage of the explosive charge during loading operations, as evidenced by withstanding a minimum internal air pressure of 85 pounds per square inch for 15 seconds without leaks when tested as specified in 4.4.3.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by defect code, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.2 First article inspection.

4.2.1 Submission.-Prior to initiating regular production, the contractor shall submit one (1) complete bomb metal parts assembly and one (1) complete set of parts, and ten (10) shipping plugs (dwg. 2518425) from each mold cavity to a Government approved facility as designated by the Contracting Officer (see 6.4) for evaluation in accordance with the provisions of 4.2.2.

The first article sample shall consist of material that has been produced by the contractor or furnished by a supplier and manufactured using the same production processes, procedures, and equipment that will be used in fulfilling the contract. The first article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply so as to significantly affect product uniformity as determined by the Government. Prior to submission the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects shall not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspections to be performed.-Each assembly and component part shall be examined for every dimension and requirement of the applicable drawing, 4.3.2 and 4.3.3.

4.2.3 Rejection.-If the assembly or any component fails to comply with any of the applicable requirements the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon observing any nonconformance of the sample to the specified requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a first article sample is accepted, the contractor is in no way authorized by the Government to initiate regular production (including procurement of materials by the contractor) unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements.

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The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and specification revisions. Changes to the process, specifications, or drawings, not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD 105. Unless otherwise approved by the contracting officer, the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest monthly quantity contractually scheduled for delivery during the contract period. Inspection lots of components or subassemblies, other than the items of delivery, shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. One agreement has been reached as to which method will be used (group or individual) that method will be used for all of the CD's listed in this specification. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4

4.3.2.1 Nose (see dwg. 9211803 covering a detail of dwg. 9226252). (If Applicable)

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical: None defined.			

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Major:	AQL 0.40 percent		
101.	Pitch diameter of thread.....	Gage	01001
Minor:	AQL 1.50 percent		
201.	Major diameter of thread, maximum (max.).....	Gage	01002
202.	Protective coating missing or inadequate.....	Visual	01003
203.	Corrosion preventive compound missing from thread.....	Visual	01004
204.	Evidence of poor workmanship (see 3.9).....	Visual	01005

4.3.2.2 Plug, closing (see dwg. 9211804 covering a detail of dwg. 9226252). (If Applicable)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Pitch diameter of thread.....	Gage	02001
Minor:	AQL 1.50 percent		
201.	Major diameter of thread, max.....	Gage	02002
202.	Protective coating missing or inadequate.....	Visual	02003
203.	Corrosion preventive compound missing from thread.....	Visual	02004
204.	Evidence of poor workmanship (see 3.9).....	Visual	02005

4.3.2.3 Plug, base (see dwg. 9233570 covering a detail of dwg. 9226252)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.50 percent		
101.	Pitch diameter of external thread.	Gage	03001
102.	Major diameter of external thread.	Gage	03002
103.	Depth of "O" ring groove.....	Gage	03003
104.	Width of "O" ring groove.....	Gage	03004
105.	Diameter of "O" ring groove.....	Gage	03005

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106.	Thickness of flange	Gage	03006
107.	Width of thread undercut, max	Gage	03007
108.	Diameter of thread undercut, min.	Gage	03008
109.	Finish on bottom of "O" ring groove improper	Visual	03009
110.	Perpendicularity of shoulder with thread	Gage	03010
111.	Thread damaged	Visual	03011
112.	Length of small inside diameter, min	Gage	03012
113.	Concentricity of adapter booster bushing seating surface with small inside diameter	Gage	03013
114.	Finish of fuze well assembly seating surface improper	Visual	03014
Minor:	AQL 2.50 percent		
201.	Holes missing	Visual	03015
202.	Anti-sieze compound missing from threads	Visual	03016
203.	Protective coating missing or inadequate	Visual	03017
204.	Evidence of poor workmanship (see 3.9)	Visual	03018
205.	Length from rear end to adapter booster bushing seating surface	Gage	03019
206.	Length of adapter booster bushing seating surface	Gage	03020

4.3.2.4 Bushing, adapter booster (see dwg. 9243971 covering a detail of dwg. 9226252)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 percent		
101.	Pitch diameter of large thread	Gage	04001
102.	Minor diameter of large thread, min	Gage	04002
103.	Pitch diameter of small thread	Gage	04003
104.	Minor diameter of small thread, max	Gage	04004
105.	Length to small threaded hole	Gage	04005

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106.	Length of outside diameter to flange	Gage	04006
107.	Location of large inside diameter.	Gage	04007
108.	Width of large inside diameter, min	Gage	04008
Minor:	AQL 2.50 percent		
201.	Blind holes missing	Visual	04009
202.	Corrosion preventive compound missing from threads	Visual	04010
203.	Protective coating missing or inaedquate	Visual	04011
204.	Evidence of poor workmanship (see 3.9)	Visual	04012

4.3.2.5 Ring, flange (see dwg. 9226255 covering a detail of dwg. 9226253).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.50 percent		
101.	Pitch diameter of thread	Gage	05001
102.	Minor diameter of thread, min	Gage	05002
103.	Perpendicularity of shoulder with thread	Gage	05003
104.	Concentricity of diameter at forward end with thread	Gage	05004
105.	Concentricity of outside diameter at rear end with thread	Gage	05005
106.	Diameter of groove pin hole or dowel pin hole, max	Gage	05006
107.	Angle of fin mounting screw groove	Gage	05007
108.	Width of fin mounting screw groove	Gage	05008
109.	Length from shoulder to end of groove	Gage	05009
110.	Diameter rear of mounting screw groove	Gage	05010
111.	Location of dowel pin hole	Gage	05011

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112.	Concentricity of "0" ring groove with thread	Gage	05012
113.	Diameter of "0" ring groove	Gage	05013
114.	Finish improper on bottom of "0" ring groove	Visual	05014
115.	Perpendicularity of base plug seating surface with thread	Gage	05015
Minor:	AQL 2.50 percent		
201.	Large outside diameter	Gage	05016
202.	Total length	Gage	05017
203.	Diameter of body seating surface, min	Gage	05018
204.	Length of body seating surface, min	Gage	05019
205.	Evidence of poor workmanship (see 3.9)	Visual	05020

4.3.2.6 Conduit (nose and tail) (see dwg. 8796749 covering a detail of dwg. 9226252).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Wall thickness	Gage	06001
102.	Rust preventive missing or inadequate on interior surface	Visual	06002
Minor:	AQL 1.00 percent		
201.	Protective coating missing or inadequate	Visual	06003
202.	Evidence of poor workmanship (see 3.9)	Visual	06004

4.3.2.7 Assembly, fuze well, nose; and assembly, fuze well, tail (see dwg. 9245939 and dwg. 9244306 covering details of dwg. 9226252).

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Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 percent		
101.	Pitch diameter of thread	Gage	07001
102.	Major diameter of thread, min	Gage	07002
103.	Inside diameter of fuze well along entire length indicated on dwg, min	Gage	07003
104.	Wall thickness, min	Gage	07004
105.	Concentricity of inside diameter with thread	Gage	07005
106.	Cracks or voids in welds	Visual	07006
107.	Thickness of flange, min (9244306)	Gage	07007
108.	Diameter of flange, min (9244306)	Gage	07008
109.	Concentricity of adaptor hole with inside diameter	Gage	07009
110.	Length to adapter	Gage	07010
Minor:	AQL 1.50 percent		
201.	Protective coating missing or inadequate	Visual	07011
202.	Evidence of poor workmanship (see 3.9)	Visual	07012
203.	Width of thread undercut, max (9244306)	Gage	07013
204.	Length of threads, min (9244306)	Gage	07014

4.3.2.8 Body (see dwg. 9226254 covering a detail of dwg. 9226253).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 percent		
101.	Wall thickness of ogival section, min. (4 readings)	Gage	08001
102.	Thickness of body through fuze well hole, min	Gage	08002
103.	Wall thickness below ogival section, min	Gage	08003
104.	Diameter of insert boss hole, max	Gage	08004

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105.	Width of combination insert boss hole, max	Gage	08005
106.	Inside diameter of body at rear end, max	Gage	08006
Minor:	AQL 1.00 percent		
201.	Large outside diameter of body	Gage	08007
202.	Evidence of poor workmanship (see 3.9)	Visual	08008

4.3.2.9 Body assembly after hydrostatic testing (see dwg. 9226253 covering a detail of dwg. 9226252).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Hydrostatic test stamp missing	Visual	09001
Major:	AQL 2.50 percent		
101.	Pitch diameter of nose thread	Gage	09002
102.	Minor diameter of nose thread, min	Gage	09003
103.	Pitch diameter of boss lug threads	Gage	09004
104.	Minor diameter of boss lug threads	Gage	09005
105.	Length of boss lug full thread, min	Gage	09006
106.	Distance between boss lug threads	Gage	09007
107.	Depth to bottom of boss lug, max	Gage	09008
108.	Pitch diameter of charging well thread	Gage	09009
109.	Concentricity of fin seating surface with nose and flange thread	Gage	09010
110.	Minor diameter of charging well thread, min	Gage	09011
111.	Length of charging well thread, min	Gage	09012
112.	Depth to bottom of charging well, max	Gage	09013
113.	Parallelism of insert bosses with centerline of bomb	Gage	09014
114.	Perpendicularity of rear end with nose thread and ring flange thread	Gage	09015

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115.	Location of set screw hole	Gage	09016
116.	Ring flange is not properly oriented	Gage	09017
117.	Weld missing or inadequate	Visual	09018
118.	Thread damaged	Visual	09019
119.	Foreign material on interior surface of bomb	Visual	09020
120.	True position of grooved pin with fin assembly seating surface	Gage	09021
Minor:	AQL 1.50 percent		
201.	Inserts not contoured with body	Visual	09022
202.	Protective coating missing or inadequate on interior of bomb	Visual	09023
203.	Evidence of poor workmanship (see 3.9)%	Visual	09024

4.3.2.10 Assembly, prior to assembling closing plug, nose plug and shipping protector (see dwg. 9226252).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Air test stamp missing	Visual	10001
Major:	AQL 1.00 percent		
101.	Torque test of nose fuze well	Test	10002
102.	Torque test of adapter booster bushing	Test	10003
103.	Concentricity of inside diameter of nose fuze well with nose thread	Gage	10004
104.	Concentricity of inside diameter of fuze well at aft end with adapter booster bushing thread	Gage	10005
105.	Inside diameter of conduit, min. (total length of conduits)	Gage	10006
106.	Inside diameter of conduits, min. (for minimum length) at nose and tail fuze well assemblies	Gage	10007
107.	Height of dowel pin or groove pin, min	Gage	10008
108.	Set screw missing	Visual	10009

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	109. Conduit above flush with bottom of charging well	Visual	10010
Minor:	AQL 1.50 percent		
	201. Retaining clip missing (if applicable)	Visual	10011
	202. Corrosion preventive compound missing from thread	Visual	10012
	203. Evidence of poor workmanship (see 3.9)	Visual	10013

4.3.2.11 Plug Shipping (see dwg. 9215093), Plug Closing Nose (see dwg. 9214994) covering a detail of dwg. 9226252).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
	101. Pitch diameter of threads, min	Gage	11001
	102. Length of thread, min	Gage	11002
Minor:	AQL 0.65 percent		
	201. Evidence of poor workmanship (see 3.9)	Visual	11003

4.3.2.12 Assembly (see dwg. 9226252).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 percent		
	101. Total length, max	Gage	12001
	102. Nose plug missing or not seated properly	Visual	12002
	103. Shipping protector missing or not secured	Visual	12003

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104.	Shipping plug missing	Visual	12004
105.	Charging well plug missing	Visual	12005
106.	Protective coating missing or incomplete (exposing base metal)	Visual	12006
Minor:	AQL 1.00 percent		
201.	Marking missing, misleading or unidentifiable	Visual	12008
202.	Evidence of poor workmanship (see 3.9)	Visual	12009

4.3.2.13 Shipping plug, nose and base, alternate (see dwg. 2518425 covering a detail of dwg. 9226252).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Pitch diameter, min	Gage	13001
102.	Major diameter, min	Gage	13002
103.	Length to flange, min	Gage	13003
104.	Total length, min	Gage	13004
Minor:	AQL 0.65 percent		
201.	Thread not complete	Visual	13005

4.3.3 Testing

4.3.3.1 First article sample

4.3.3.1.1 Strength of weld

4.3.3.1.1.1 Nose sleeve weld, Code No. 07015

4.3.3.1.1.2 Nose adapter weld, Code No. 07016

4.3.3.1.1.3 Tail sleeve weld, Code No. 07017

4.3.3.1.1.4 Tail adapter weld, Code No. 07018

4.3.3.1.2 Internal hydrostatic pressure of bomb body, (see 3.4) Code No. 09025.

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4.3.3.1.3 Brinnel hardness of bomb body (see dwg. 9226253) Code No. 09026.

4.3.3.1.4 Surface preparation prior to painting (see 3.7) Code No. 09027.

4.3.3.1.5 Seal of fuzing compartments (see 3.11 and 4.4.3) Code No. 10017.

4.3.3.1.6 Integrity of circumferential tail weld (see 3.6.3 and 4.4.6) Code No. 09028.

4.3.3.1.7 Depth of penetration of circumferential tail weld (see dwg. 9226252) Code No. 09029.-The portion of the first article sample representing the weld from each welding machine shall be tested as specified in 4.4.7.

4.3.3.2 Regular production

4.3.3.2.1 Testing.-The sampling plans for the following characteristics are designed specifically for the required reliability levels which are not attainable by the use of Standards MIL-STD-105 and MIL-STD-1235. Inspection equipment necessary for the performance of these tests shall be in accordance with 4.3.4.

4.3.3.2.2 Strength of fuze well assembly welds (see 3.3) Major defects: nose sleeve weld, Code No. 07015; nose adapter weld, Code No. 07016; tail sleeve weld, Code No. 07017; and tail adapter weld, Code No. 07018.-A sample of two (2) fuze well assemblies shall be selected from each welding machine at the start, midpoint and end of each shift. The sample shall be tested as specified in 4.4.1, using the test equipment specified in 4.3.4. If one or both assemblies are classed defective, all assemblies produced by the welding machine responsible for the defective assembly shall be tested 100 percent. This 100 percent test shall be performed as specified in 4.4.1, and defective assemblies shall be rejected and removed from the lot. The welding machine producing the defective weld shall not be permitted to continue production until corrective fuze well assemblies are produced, tested and found to be acceptable.

4.3.3.2.3 Internal hydrostatic pressure of bomb body (see 3.4) Critical defect, Code No. 09025.-This test shall be performed 100 percent. The test shall be performed as specified in 4.4.2, using test equipment specified in 4.3.4. Any assembly which fails to comply with the specified requirement shall be classed defective and removed from the lot.

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4.3.3.2.4 Brinnel hardness of bomb body (see dwg. 9226253) - Major defect, Code No. 09026.-This test shall be performed 100 percent. Any bomb body which fails to comply with the specified requirement shall be classed defective and removed from the lot. Testing shall be in accordance with Method 242.1 of Federal Test Method Standard No. 151.

4.3.3.2.5 Surface preparation prior to painting (see 3.7) - Major defect, Code No. 09027.-Nine units shall be selected from each 8 hours production for this determination. Three each shall be selected at the start, mid-point and end of each work shift. If one or more units of a periodic sample fail in meeting one or more of the requirements all units cleaned since the last acceptable test shall be rejected subject to recleaning and retest.

4.3.3.2.6 Seal of fuzing compartments (see 3.11) Critical defect, Code No. 10017.-This test shall be performed 100 percent. This test shall be performed on parts assemblies prior to assembly of noses and adapter booster bushings. This test shall be performed as specified in 4.4.3 using the test equipment specified in 4.3.4. Defective assemblies shall be rejected and removed from the lot.

4.3.3.2.7 Depth of "O" ring groove (see dwg. 9226255) Major defect, Code No. 05021.-This inspection shall be performed 100 percent, using test equipment specified in 4.3.4. Any flange ring which fails to comply with the specified drawing requirement shall be classed defective and removed from the lot.

4.3.3.2.8 Bushing adapter-booster above flush with aft face of base plug (see dwg. 0226252) Major defect, Code No. 10018.-This inspection shall be performed 100 percent, using test equipment specified in 4.3.4 Any assembly which fails to comply with the specified requirement shall be classed defective and removed from the lot.

4.3.3.2.9 Base plug above flush with aft surface of ring flange (see dwg. 9226252) Major defect, Code No. 10019.-This inspection shall be performed 100 percent, using test equipment specified in 4.3.4.

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Any assembly which fails to comply with the specified requirement shall be classed defective and removed from the lot.

4.3.3.2.10 Stress crack test on the shipping protector ring (see 3.10) Major defect, Code No. 12010.-The sampling plan for this test shall be in accordance with Standard MIL-STD-105, using applicable sample size code letter under special inspection level S4, in accordance with Table I at an AQL of 4.00 percent, using single sample plan under Table II-A. Any shipping protector ring that shows evidence of breaks or cracks shall be classed defective.

4.3.3.2.11 Integrity of circumferential tail weld (see 3.6.3) Major defect, Code No. 09028.-This test shall be performed 100 percent. The test shall be performed as specified in 4.4.6 using the test equipment specified in 4.3.4. Welded bodies classed defective shall be rejected and removed from the lot.

4.3.3.2.12 Depth of penetration of circumferential tail weld (see dwg. 9226252) Major defect. Code No. 09029.-One (1) welded bomb body, after all heat treatment operations and hydro-test, shall be selected from each week's production (one bomb from the entire contractor's production for the week) and tested as specified in 4.4.7. If the sample is classed defective, one (1) additional welded bomb body shall be selected from each welding machine (after all heat treatment operations and hydro-test) and tested as specified in 4.4.7. Welding machines producing unacceptable welds shall be removed from production and the necessary adjustments and repairs made, and shall not return to production until a satisfactory weld is produced as evidenced by testing as specified in 4.4.7.

4.3.4 Inspection Equipment.-Equipment Tabulation Number ET-9226252 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

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4.3.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed in the ET shall be in accordance with Specification MIL-I-45607.

4.3.4.3 Government use of contractor's inspection equipment.-The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3.4.4 Calibration.-The contractor shall have available a set of master gages, standards, and appropriate instruments for regularly scheduled calibrations of his acceptance inspection equipment. Records of such regularly scheduled calibrations shall be maintained by the contractor and made available for review by the Government. The calibration of gages, standards, and instruments shall be in accordance with Specification MIL-C-45662.

4.4 Test methods and procedures.-The test method and procedures described in each of the following tests represents only one method and one procedure which has been proven in use and is approved by the Government for Government acceptance purposes. Producers and manufacturers have the option of using substitute methods and procedures only after obtaining Government approval. Government approval will be based on an evaluation to determine that the substitute method and procedure will result an equal or better quality evaluation than that specified herein.

4.4.1 Strength of fuze well assembly welds. The fixture for the sleeve weld test shall consist of a ring to support the fuze well assembly at the sleeve and the specified load applied to the bottom of the fuze well (inside bottom) by means of a cylindrical ram. The ram should be contoured to clear the adapter, and conform to the geometry of the bottom of the fuze well and should be cushioned to prevent damage to the finish of the fuze well assembly. The adapter weld test fixture shall consist of a contoured ring, cushioned to prevent damage to the finish of the fuze well, and shall support the fuze well assembly on the outside bottom of the fuze well and the required load applied to the adapter from the inside end of the adapter. The required load shall be maintained for a minimum of two (2) seconds. After removal of the assembly from the test fixture, the fuze well assembly shall be examined as specified in 4.3.2.7. The fuze well assembly shall be classed defective if it becomes disassembled, deforms beyond the drawing requirements, or if it ruptures or exhibits cracked welds.

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4.4.2 Internal hydrostatic pressure of bomb body.-One end of the body assembly and the charging well shall be sealed prior to test. The hydraulic media shall be introduced at the unsealed end of the bomb body by means of an adapter which will isolate the pressure source, then the desired pressure shall be introduced into the bomb body. After the specified pressure has been attained, the pressure supply shall be cut off and held for a minimum of 10 seconds. Any assembly which shows evidence of leaks, cracks or rupture shall be classed defective. A steady stream of air bubbles shall consititute a leaker (if applicable). If the unit is found acceptable, the letter "H" shall be stamped in accordance with the applicable drawing.

4.4.3 Seal of fuzing compartments.-The metal parts assembly, prior to assembly of the nose and adapter booster bushing, shall be prepared for testing by sealing the threaded end of both fuze wells and installing a fixture which will seal the charging well (but not the joint between the conduits and the charging well) and provide a device for introducing the required air pressure. The ancillary equipment shall be capable of pressurizing the conduits and fuze wells to the specified internal air pressure, shall be capable of isolating the conduit and fuze wells after pressurization and monitoring the level of pressure after isolation. If the unit meets the specified requirement of 3.11 letter "A" shall be stamped thereon in accordance with the applicable drawing.

4.4.4 Surface preparation prior to painting.-The components or assemblies shall be tested as specified in TT-C-490.

4.4.5 Stress crack test.-The shipping protector shall be cut into sections so that the weld line (see 6.5) is encountered in each section. The individual sections shall then be placed in a three finger clamp (with asbestos sleeves over each of the fingers) at the weld line.

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Two fingers shall be closed on the outside circumference of the ring section, and the single finger shall be closed on the inside circumference of the ring section. The clamp shall then be tightened until the fingers are approximately 2 inches apart. The sample shall then be submerged into a large mouth beaker or a pail of pure methanol at room temperature for 30 plus or minus 2 seconds, then removed and inspected for breaks or cracks in the material. Dry out clamp before repeating process (use a dryer).

4.4.6 Integrity of circumferential tail weld.-The probe of the specified ultra-sonic device shall be moved as specified in the test equipment operational instructions to scan the entire 360 degrees of weld. The extent of discontinuities shall be marked on the surface of the bomb flange that is in contact with the probe. These marks shall be radial lines with marking to show between which lines the discontinuities exist. The length of the discontinuity shall be measured using a flexible scale bent along the contour of the raised portion of the flange adjacent to the surface that was in contact with the probe. The welded bomb body shall be classed defective if it fails to conform to the specified requirements.

4.4.7 Depth of penetration of circumferential tail weld.-The welded bomb body shall be cut longitudinally to expose four approx. equally spaced surfaces, one of which shall contain the first weld overlap. These surfaces shall be polished and etched to provide adequate contrast between weld and base metal. The bomb body shall be classed defective if one or more surfaces indicate inadequate weld penetration.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packing - NOT APPLICABLE

5.2 Marking.-Marking shall be as specified in Drawing 9226252.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

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6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists.-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

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6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs, All submitted designs shall contain a reference to the applicable 5 digit Code Number contained in section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-M. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Submission of First Article Samples.-Instructions as to the location for the evaluation of First Article Sample shall be obtained from the contracting officer. Upon receipt of such request, the contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries shall be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-M, Dover, New Jersey 07801.

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6.5 Weld line.-Weld line is defined as the area between the two gates. Gates are small projections on the ring which can be seen and are located where the plastic flows into the mold to make the part.

Custodian:
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Preparing Activity:
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